Sylvester (Siyuan) Zhang

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Education

University of Michigan August 2026 ~ Now
PhD in Robotics *Department of Robotics*

- Advisor: Prof. Camreon Aubin & Prof. Talia Moore

Columbia University August 2024 ~ August 2026
MS in Mechanical Engineering (Robotics and Control) *Department of Mechanical Engineering*

- **Advanced Master Research Program**, Advisor: Prof. Hod Lipson

Sun Yat-sen University August 2020 ~ June 2024
BE in Aeronautical and Astronautical Engineering *School of Aeronautics and Astronautics*

Research Interest

Embodied Intelligence, Soft and Bio-inspired Robotics, Reconfigurable and Modular Systems, Human-centered Assistive Robot, Robotic Material, Computational Design, Robot Learning

Publication

- **Zhang, S.**, **Lipson, H.** (2026). Helical Genesis: An Intelligent Self-Reproducing Robot Evolving from 2-D lattices into 3-D DNA-Inspired Architectures. *Manuscript in Preparation.*
- Zhang, Y.*, **Zhang, S.***, **Lipson, H.** (2026). Design and Development of a Human-like Lip-Sync Robot Mechanism and Control Strategy. *Manuscript in Preparation.* (*Co-first authors)
- **Zhang, S.**, Zhang, Y., Lyu, J., & **Agrawal, S. K.** (2026). From Structural Design to Dynamics Modeling: Control-Oriented Development of a 3-RRR Parallel Ankle Rehabilitation Robot. *Manuscript in Preparation.*
- Zhao, Y., Zhu, J., Zhang, J., **Zhang, S.**, Shao, M., Chai, Z., ... & **Zhang, J.** (2025). Enhancing Grasping Diversity with a Pinch-Suction and Soft-Rigid Hybrid Multimodal Gripper. ***IEEE Transactions on Robotics***.
- Ma, K., Zhang, J., Sun, R., Chang, B., **Zhang, S.**, Wang, X., ... & **Zhang, J.** (2024). Synergizing Structural Stiffness Regulation with Compliance Contact Stiffness: Bioinspired Soft Stimuli-Responsive Materials Design for Soft Machines. ***Advanced Engineering Materials***, *26*(10), 2400461.
- Zhao, Y., Zhang, J., **Zhang, S.**, Zhang, P., Dong, G., **Wu, J.**, & **Zhang, J.** (2023). Transporting dispersed cylindrical granules: An intelligent strategy inspired by an elephant trunk. ***Advanced Intelligent Systems***, *5*(10), 2300182.

Research Experience

Design of An Intelligent Self-Reproducing Robot Jan 2025 ~ Present
Advisor: Hod Lipson *Columbia University*

- Developed a magnetically coupled, servo-actuated triangular module system capable of autonomous self-reproduction under positional uncertainty, achieving stable 3D–2D shape morphing and tetrahedral assembly.
- Engineered a new reproduction strategy reducing the minimal replicating unit from 4-module to 2-module "cells," enabling multi-generation replication and robust attachment/detachment through a distributed communication protocol with minimal sensing.

Development of a 3-RRR Parallel Ankle Rehabilitation Robot Jan 2025 ~ Present
Advisor: Sunil K. Agrawal *Columbia University*

- Developing a wearable 3-RRR parallel mechanism to improve portability and motion freedom in ankle rehabilitation robot compared with traditional design.
- Designed and simulated inverse kinematics and Jacobian-based control, enabling precise motion tracking for multi-axis ankle support. Integrating therapist-inspired assistance/resistance modes, currently under validation for rehabilitation.

Pinch-Suction and Soft-Rigid Hybrid Multimodal Gripper Apr 2023 ~ Apr 2024
Advisor: Prof. Jianing Wu, Prof. Jinxiu Zhang *Sun Yat-sen University*

- Designed a soft-rigid hybrid gripper with multiple manipulation modes, enabling effective handling of objects of varying sizes and weights to address the limited adaptability of conventional grippers.
- Integrated gripping and suction mechanisms to increase stability and controllability during object manipulation.

- Enabled grasping diversity across weight (0.2 g–10 kg), fragility (jelly to aluminum), scale (0.46 mm–0.55 m), and shape (poorly pinchable/sackable), outperforming conventional grippers.

Bioinspired Stiffness-Regulating Materials for Soft Machines Jun 2023 ~ Jan 2024

Advisor: Prof. Jianing Wu, Prof. Jinxiu Zhang *Sun Yat-sen University*

- Developed elastomer-SMA composite structures with Joule-heat stiffness regulation to address the compliance-load trade-off in soft robotics.
- Characterized dynamic stress-strain responses and rapid stiffness switching through systematic material testing.
- Applied the material to adaptive grippers and wearable prototypes, enabling enhanced load bearing and versatility.

Elephant Trunk Inspired Soft Gripper May 2022 ~ Jun 2023

Advisor: Prof. Jianing Wu, Prof. Jinxiu Zhang *Sun Yat-sen University*

- Proposed and prototyped a pneumatic gripper inspired by elephant trunk mechanics to tackle the challenge of manipulating irregular and granular object.
- Optimized actuation pressures and deformation profiles through ANSYS and SOLIDWORKS simulations.
- Achieved >90% grasp success rate and 50% faster operation time compared to baseline grippers in experimental validation.

Project experience

Soft Robotic & Neural Interface Material Aug 2025 ~ Dec 2025

Advisor: Prof. Chaoqun Dong *Columbia University*

- Performing finite element simulations of soft neural interface devices, investigating strain localization, viscoelastic deformation, and interfacial adhesion to guide structural optimization of stretchable robotic materials.
- Designed and fabricated customized experimental platforms and mounting fixtures to improve the reproducibility and precision of soft-material testing.

Design and Gait Optimization of a Quadruped Spider Robot for Robotics Studio Sep 2024 ~ Dec 2024

Advisor: Prof. Hod Lipson *Columbia University*

- Designed a 4-legged robotic system with 2 DOF per leg, powered by 8 servomotors (240° range) and controlled via a Raspberry Pi. Created a parametric CAD model in SOLIDWORKS, converted it to a URDF file, and optimized the robot's gait using PyBullet simulations and parallel hill climber algorithms.
- Implemented control algorithms in Python to actuate servos, enabling sim-to-real to optimize walking patterns.
- Built and tested the prototype operating 3D printing for rapid iterations, attaining a final walking speed of 29 cm/s.

Design of a Bionic Flapping-Wing Robot Inspired by Birds Sep 2022 ~ Dec 2023

Advisor: Prof. Zhenbo Lu *Sun Yat-sen University*

- Developed a lightweight, foldable, and low-noise flapping-wing robot inspired by small birds, integrating principles of bionics.
- Devised and simulated the mechanical model using SOLIDWORKS and ANSYS, modifying structural performance and aerodynamics.
- Implemented a closed-loop control system to achieve stable and controlled flight.

Design of a Foldable, Adhesive Crawling CubeSat for Space Station Operations Mar 2022 ~ Aug 2023

Advisor: Prof. Jinxiu Zhang, Jianing Wu *Sun Yat-sen University*

- Led a team of 5 in designing a spider-inspired CubeSat with foldable claws, mechanical legs, and detection modules for surface inspection, debris removal, and damage repair.
- Optimized mechanical structure through SOLIDWORKS and explored electrostatic adsorption for secure attachment in space environments.
- Attained 3rd prize in the IAF-CSA Space Universities CubeSat Challenge 2.0 for innovative design and engineering solutions.

Teaching Experience

Graduate Teaching Assistant | Columbia University 2026 Spring

MECE 4058 mechatronics & Embedded Microcomputer Control Advisor: Enrico Zordan

- Supporting 20+ students across undergraduate and graduate levels through mechatronics experiments and lab sessions.
- Mentored hardware/software integration projects spanning DC and stepper motors, electromagnetic levitation, and closed-loop control.

Graduate Teaching Assistant | Columbia University 2025 Spring
MECE 4611 Robotics Studio Advisor: Hod Lipson

- Contributed to robotics teaching and curriculum development, integrating python algorithm/ embedded systems to support 90+ students in hands-on robotic projects across undergraduate and graduate levels.
- Provided technical support and debugging help while maintaining positive student feedback across course evaluations.

Professional Experience

Amazon May 2026 ~ Aug 2026
Product Design Engineer Intern

- Designed and iterated acoustic enclosure components in CAD (PTC Creo) across three full design cycles, rebuilding a fragile inherited model into a simpler, more robust structure and converging on a manufacturable, interference-free assembly within a tightly packed, space-constrained device cavity.
- Conducted engineering trade-off analysis, balancing component sizing against thermal, sealing, and packaging constraints, and ran interference/fit checks against surrounding electronics to guarantee zero collisions; delivered a validated design handed off for 3D-printed prototyping.
- Reverse-engineered a multi-link geared hinge mechanism through teardown analysis, then designed an original compact hinge assembly from the ground up, applying GD&T principles (datum referencing, clearance fits) to achieve smooth motion, controlled torque feel, and durability, and integrated it into a full assembly in CAD.
- Took ownership of a project mid-stream with no prior context and drove it independently from concept to validated design, while collaborating with cross-functional teams (advanced materials, audio, mechanical engineering) across time zones through regular design reviews.

Insta360 Jan 2024 ~ Mar 2024
Structural Design Engineer Intern

- Contributed to the Insta360 X4 thermal accessories R&D program, developing structural concepts in CAD and iterating component geometry against thermal dissipation and compact packaging requirements to support stable performance during extended camera operation.
- Designed and refined the Insta360 Ace Pro 2 waterproof lens protection cover, balancing sealing performance, optical clearance, and user-facing fit, and validated the design through fit checks against the surrounding lens assembly.
- Produced detailed manufacturing-ready technical drawings with tolerancing and material callouts, ensuring the designs met production and assembly requirements for handoff to downstream fabrication teams.
- Drove design optimization for thermal management and protective functionality, evaluating trade-offs across structural robustness, weight, and manufacturability to converge on reliable, cost-effective accessory solutions.

Academic Service

- Reviewer of NeurIPS

Technical Skills

- Programming: Python, C, MATLAB, C++, ROS, Linux, Arduino IDE
- Software: Auto CAD, MATLAB, ANSYS, SOLIDWORKS, PTC Creo, CATIA, NX CAD
- Engineering Techniques: FEA, DFM, DFA, FA, DOE, GD&T
- Manufacturing Techniques: FDM 3D Printing, Stereolithography (SLA), Laser Cutting, CNC Machining